Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

Code: CH-PICU-PRT-14-Vers.1.0

Effective Date: 12/10/2023
Target Review Date: 10/10/2026

Title: Assessment and Management of Delirium in PICU, PCSU and PHDU

Assessment and Management of Delirium in PICU, PCSU and PHDU

Pediatric delirium is a common diagnosis, though under recognized and under studied phenomenon in pediatric critical care.

The brain can react to illness, particularly critical illness, with several responses of which the more important are sickness behavior, fever, epilepsy, catatonia, delirium, refractory agitation, and coma.American Psychiatric Association has recommended that “delirium” be consistently used to describe the “acute state of brain dysfunction” in critical care literature.

There are certain risk factors needs to be considered for patients who are prone to develop Delirium:

I WATCH DEATH

I	**Infections**	Encephalitis, meningitis, urinary tract infection, pneumonia…
W	**Withdrawal**	Alcohol, barbiturates, benzodiazepines…
A	**Acute metabolic**	Electrolyte imbalance, hepatic or renal failure…
T	**Trauma**	Head injury, postoperative…
C	**CNS pathology**	Stroke, haemorrhage, tumour, seizure disorder…
H	**Hypoxia**	Anaemia, cardiac failure, pulmonary embolus…
D	**Deficiencies**	Vitamin B12, folic acid, thiamine…
E	**Endocrinopathies**	Thyroid, glucose, parathyroid, adrenal…
A	**Acute vascular**	Shock, vasculitis, hypertensive encephalopathy…
T	**Toxic or drugs**	Toxins, substance intoxication, medications (alcohol, anaesthetics, anticholinergics, narcotics etc.)
H	**Heavy metals**	Arsenic, lead, mercury…

Delirium is classified as:

- Hyperactive delirium (Agitated, irritable and thrash about).
- Hypoactive delirium (Apathetic, uninterested).
- Mixed delirium (Hyperactive and hypoactive).
- Emergence delirium (Benign course and usually resolves completely in 30 to 45 minutes).

The Pediatric Confusion Assessment Method for the Intensive Care Unit (pCAM-ICU) was the first validated tool for objective delirium monitoring in critically ill children over 5 years of age, demonstrating a specificity of 99% and sensitivity of 83%. Later Cornell Assessment for Pediatric Delirium (CAPD) was introduced as a subjective delirium screening tool in critically ill children with a reported specificity of 79% and sensitivity of 94%.

By educating and supporting staff nurses with this new protocol process, we expect to:

(1) Decrease the risk factors for developing delirium.

(2) Identify symptoms and behaviors of delirium early in the hospital stay.

(3) Decrease length of stay in the PICU, and

(4) Improve patients' outcome where a diagnosis of delirium is present.

In many centres it is also recognized:

(1) Increase in misdiagnosis of agitation versus a delirium diagnosis.

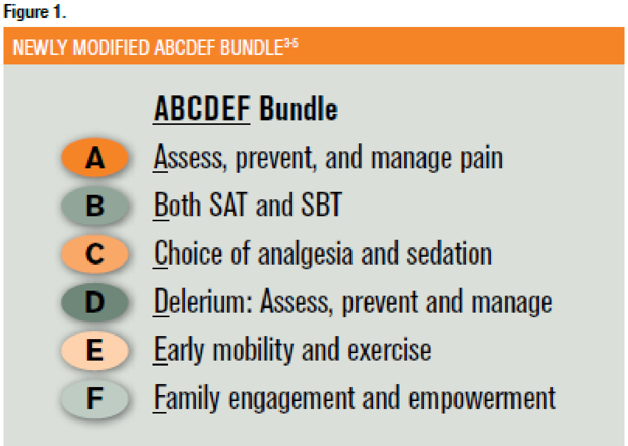
(2) Increase in use of benzodiazepines for sedation without regard to length of use.

(3) Deficit in knowledge about pediatric delirium in nurses, and

(4) Need for a standardized care plan to treat patients with delirium in PICU.

Initial step is to prevent delirium by applying ABCDEF Bundle:

Figure 1.



SAT = spontaneous awakening trial; SBT = spontaneous breathing trial

SAT: Spontaneous Awakening Trial, SBT Spontaneous Breathing Trial

Since the condition is not well studied and under recognized in our unit, we will screen all patients admitted in PICU for delirium once weaning from sedation started. The test can be started while patient is ventilated.

The tool for screening is divided as below:

SBS Score	Description
+2	Agitated
+1	Restless and difficult to calm
0	Awake and able to calm
-1	Responsive to gentle touch or voice
-2	Responsive to noxious stimuli
-3	Unresponsive

If score more than -2 proceed to next step CAPD Assessment for children

Cornell Assessment Pediatric Delirium (CAPD) for children

SBS Score (if -2 or -3 do not proceed)

Please answer the following questions based on your interaction with the patient over the course of your shift:

	Never 4	Rarely 3	Sometimes 2	Often 1	Always 0	Score

1. Are the child's actions purposeful?						
1. Is the child aware of his/her surroundings?						
4.Does the child communicate needs and wants?						
	Never 0	Rarely 1	Sometimes 2	Often 3	Always 4	Score
5.Is the child restless?						
6.Is the child inconsolable?						
7.Is the child underactive – very little movement while awake?						
8.Does it take the child a long time to respond to interactions?						

If CAPD score is > 9 it is delirium

Developmental Anchor Points For Youngest Patients

	NB	4 weeks	6 weeks	8 weeks	28 weeks	1 year	2 years
1. Does the child make eye contact with the caregiver?	Fixates on face	Holds gaze briefly Follows 90 degrees	Holds gaze	Follows moving object/caregiver past midline, regards examiner's hand holding object, focused attention	Holds gaze. Prefers primary parent. Looks at speaker	Holds gaze. Prefers primary parent. Looks at speaker	Holds gaze. Prefers primary parent. Looks at speaker
2. Are the child's actions purposeful?	Moves head to side, dominated by primitive reflexes	Reaches (with some discoordination)	Reaches	Symmetric movements, will passively grasp handed object	Reaches with coordinated smooth movement	Reaches and manipulates objects, tries to change position, if mobile may try to get up	Reaches and manipulates objects, tries to change position, if mobile may try to get up and walk
3. Is the child aware of his/her surroundings?	Calm awake time	Awake alert time Turns to primary caretaker's voice May turn to smell of primary care taker	Increasing awake alert time Turns to primary caretaker's voice May turn to smell of primary care taker	Facial brightening or smile in response to nodding head, frown to bell, coos	Strongly prefers mother, then other familiars. Differentiates between novel and familiar objects	Prefers primary parent, then other familiars, upset when separated from preferred care takers. Comforted by familiar objects especially favorite blanket or stuffed animal	Prefers primary parent, then other familiars, upset when separated from preferred care takers. Comforted by familiar objects especially favorite blanket or stuffed animal
4. Does the child communicate needs and wants?	Cries when hungry or uncomfortable	Cries when hungry or uncomfortable	Cries when hungry or uncomfortable	Cries when hungry or uncomfortable	Vocalizes /indicates about needs, eg. hunger, discomfort, curiosity in objects, or surroundings	Uses single words, or signs	3-4 word sentences, or signs. May indicate toilet needs, calls self or me
5. Is the child restless?	No sustained awake alert state	No sustained calm state	No sustained calm state	No sustained calm state	No sustained calm state	No sustained calm state	No sustained calm state
6. Is the child inconsolable?	Not soothed by parental rocking, singing, feeding, comforting actions	Not soothed by parental rocking, singing, feeding, comforting actions	Not soothed by parental rocking, singing, feeding, comforting actions	Not soothed by parental rocking, singing, comforting actions	Not soothed by usual methods eg. singing, holding, talking	Not soothed by usual methods eg. singing, holding, talking, reading	Not soothed by usual methods eg. singing, holding, talking, reading (May tantrum, but can organize)
7. Is the child underactive— very little movement while awake?	Little if any flexed and then relaxed state with primitive reflexes (Child should be sleeping comfortably most of the time)	Little if any reaching, kicking, grasping (still may be somewhat discoordinated)	Little if any reaching, kicking, grasping (may begin to be more coordinated)	Little if any purposive grasping, control of head and arm movements, such as pushing things that are noxious away	Little if any reaching, grasping, moving around in bed, pushing things away	Little if any play, efforts to sit up, pull up, and if mobile crawl or walk around	Little if any more elaborate play, efforts to sit up and move around, and if able to stand, walk, or jump
8. Does it take the child a long time to respond to interactions?	Not making sounds or reflexes active as expected (grasp, suck, moro)	Not making sounds or reflexes active as expected (grasp, suck, moro)	Not kicking or crying with noxious stimuli	Not cooing, smiling, or focusing gaze in response to interactions	Not babbling or smiling/laughing in social interactions (or even actively rejecting an interaction)	Not following simple directions. If verbal, not engaging in simple dialogue with words or jargon	Not following 1-2 step simple commands. If verbal, not engaging in more complex dialogue

There are several common causes of delirium. After completing the history and medical assessment, the clinical team should consider these causes and recommendations using the BRAIN MAPS acronym:

	Assessment	Evaluation	Recommendations

B	**Bring Oxygen**	• Evaluate for:
		  ◦ Hypoxemia
		  ◦ Low cardiac output
		  ◦ Anemia
			• Improve oxygenation via:
			  ◦ O_2 delivery
			  ◦ Resolution of anemia (PRBCs)

R	**Remove/Reduce Drugs**	• Evaluate for use of Anticholinergics and sedative medications	• Discontinue if possible
A	**Atmosphere**	• Room setup — lights, noise levels • Restraint use • Caregiver presence • Schedule/routine • Use of adaptive equipment and/or communication aids (e.g. glasses/hearing aids)	• Encourage normal day/night routine • Encourage consistent and familiar caregiver presence
I	**Infection/Mobilization/Inflammation**	• Infectious workup	• Treat infection and fever

N
M
	New organ dysfunction
	and
	Metabolic disturbance
		• Consider all system: CNS, CV, pulmonary, hepatic, renal, endocrine
		• Evaluate with CMP and ABG for:
		  ◦ Hypo/hypernatremia
		  ◦ Hypo/hyperkalemia
		  ◦ Hypocalcemia
		  ◦ Alkalosis/acidosis
			• Normalize electrolytes
			• See information below on Emergence Agitation and NMDA Encephalitis

A	**Awake**	• No bedtime routine • Sleep wake cycle disturbance	• Establish day/night cycles
P	**Pain**	• Untreated or undertreated pain • Over-treated (sedated)	• Adjust analgesia regimen if appropriate
S	**Sedation**	Critically evaluate all benzodiazepine use Set sedation target	• Consider weaning or discontinuing benzodiazepines • Consider adding Dexmedetomidine in patients with appropriate hemodynamics.

Differentiating sedation withdrawal from delirium in patient on weaning dose of opioids/

Benzodiazepines can be challenging:

-Delirium can be present from day1 of illness whereas withdrawal during weaning phase of sedation.

-**Delirium** is altered state of alertness with disorganized content. Eg patient look alert, but not focusing or interacting with parents as expected or agitated with autonomic dysfunction.

-**Withdrawal** is characterized by autonomic (fever, tachycardia, tachypnea, sweating and tremors, diarrhea etc) and hyper stimulus state. WAT-1 is excellent tool to diagnosis it. Alertness and content of conscious ness are usually normal in withdrawal.

Treatment:

Step 1: Non Pharmacological

Hypoactive Delirium: (A child with hypoactive delirium may have a negative RASS score even in the absence of pharmacologic sedation).

1. A.Cognitive stimulation (Introduce T.V, Play Qur'an, or favorite music, show favorite cartoons, use iPad, rattles for infants, encourage parents to bring favorite toys from home, keep under sunlight and normal lights).
2. Patient reorientation by engaging parents.
3. Consistent non pharmacologic sleep.
4. Exercise and early mobilization.
5. Timely removal of indwelling catheters.
6. Early parents' engagement with the child.

Hyperactive Delirium:

a.Cognitive inhibition (If possible keep in single room, with diurnal effect eg; switch offlights at night and switch on at morning to keep the sleep awake cycle regulated, try to have as less noise as possible).

b.Patient reorientation by engaging parents.

c.Consistent non pharmacologic sleep.

d.Exercise and early mobilization.

e.Timely removal of indwelling catheters.

f.Early parents' engagement with the child.

Mixed Delirium:

As per type of delirium at a time, if behaving hypo or hypo follow the management plan respectively as noted above.

Emergent delirium:

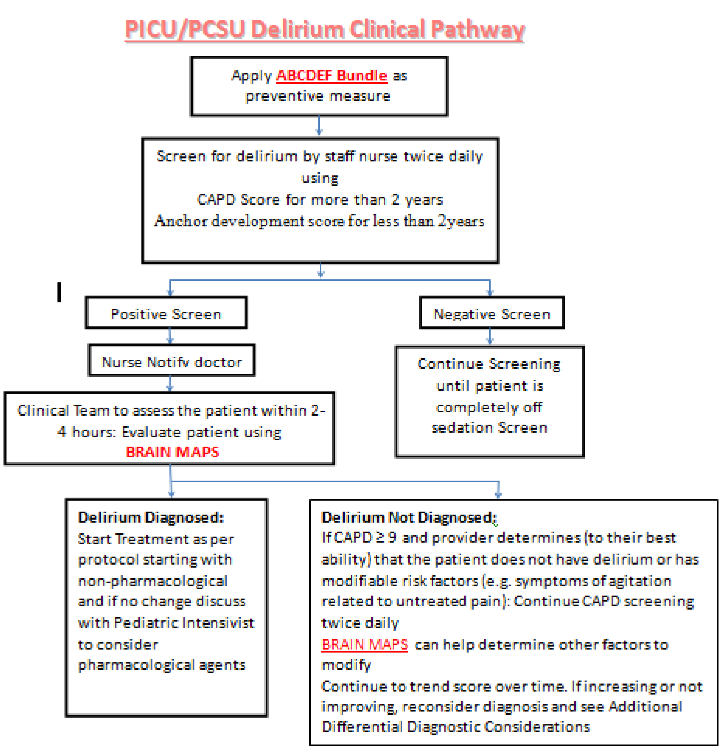
Just needs time and engage parents early with child.

If non pharmacological measures fail proceed to:

Step 2: Pharmacological:

1.Haloperidol (Haldol) loading dose of 0.15 to 0.25 mg intravenous followed by 0.05 to 0.5 mg/kg every day for maintenance until sign and symptoms disappears.

2.In patients who can tolerate oral medication, risperidone (Risperdal), as a loading dose of 0.1 to 0.2 mg followed by 0.2 to 2.0 mg every day as maintenance until sign and symptoms disappears.



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Review Details

	Review BY	Review Date
1	PICU Team	17/10/2019

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